

Problem 4

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Problem 1 Two polynomials, h and g , are given

$$h(x) = \frac{x^2 - 4}{4}$$

$$g(x) = x - 2$$

State the form of the limit, evaluate the limit, or state that it does not exist.
Justify your answer.

(a) $\lim_{x \rightarrow 2} \frac{h(x)}{g(x)}$

(b) $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3}$

(c) $\lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4}$